

Ink Morrow

WHERE STORIES GROW CLAWS

5.0 EDITION

Maintainer, Testing & Release Handbook

Change playable fiction without breaking history or trust

For contributors and release owners / November 2026

Maintainer, Testing & Release Handbook

Change the playable-fiction product without losing causal history, paid authority or honest evidence. This handbook describes current development and release practice, not the retired writing-suite workflow.

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01 Narrative qualification and its limits

N1–N6 adds read-only source/context projections, complete-character and next-move guidance, existing-call reference authoring guidance, and a bounded opt-in rolling evaluation protocol. No schema or accepted-fiction rewrite is involved. Current runtime/formats are the target; the owner waived backwards compatibility.

Deterministic tests cover exact old evidence beyond the recent window, complete reference fields, ownership/knowledge separation, local assertion versus intention, relationship-target relevance, alternate-path/future isolation, prior refusal then new sufficient grounds, empty-next-chapter handoff, quiet Follow with three promises, current-format source remapping and capacity rejection before purchase. Reference creation checks seed preservation, one-call replay and no automatic template/story adoption. Existing narration/speech/ownership/quality contracts remain covered.

The separate NARRATIVE-EVALUATION-PROTOCOL.md fixes cases, source/fixture/model settings, rolling chapters, budgets, retained failures, contradiction/repeated-remedy/ initiative/payoff criteria and evaluator disagreement before observation. It is a proposal, not authorization or a completed benchmark. No paid trial was run. Report **Live narrative quality not evaluated** beside mechanical CI results; mocked replies and instruction-presence checks do not prove model behavior or superiority.

02 B7 readable books, Bookshelf and reading copies

Gate separates **Make a readable book** from **Keep playing elsewhere**. A book uses the viewed path and inspection moment without activating it. Set title, author credit, language and optional front/back matter. Details are private server drafts and recover after reload; edits invalidate the current preview. Preview shows contents, included cover/placed art, sample prose and exclusions. Empty planned groups, Ask, directions, private references, other paths and audio do not enter the frozen PublicationDocument. Prose itself may contain fictional secrets.

Build frozen edition renders selected formats or all nine locally: DOCX, ODT, RTF, EPUB, PDF, HTML, Markdown, plain text and JSON. Every adapter uses the same immutable source. EPUB places art on a separate page immediately before its prose. A build is a durable job, independent of later story changes. Reopen a retained edition to inspect/cancel a running job. Restart marks abandoned jobs interrupted; navigation/restart never silently rebuilds. A partial failure retains successful formats. Missing/damaged files remain unavailable until an explicit local build; verified identical files can be reused for free. Earlier editions stay frozen.

Bookshelf loads bounded pages of retained books, audio and owner-only story art. Filter the collection, download available formats, play prepared MP3, or return to the source story. Older source, interrupted jobs and unavailable bytes are labelled. Loading the collection starts no renderer, model request or full-story download. Removing a book edition revokes its links and releases only files with no other edition pins. Source fiction remains intact. Private `.inkmorrow5` saves preserve all paths and source images, but exclude prepared books/audio, jobs and reading-link authority. Download prepared editions separately.

Reading links appear only when an administrator configures `PUBLIC_SHARE_BASE_URL` as an HTTP(S) origin and separately provides reachable hosting/allowed-host/proxy configuration. Local export works with sharing unconfigured. Review the exact frozen contents, choose expiry and explicitly create a link. Anyone possessing it can read that copy until expiry/revocation; downloaded copies cannot be retracted. Copy the URL, or select it manually if clipboard access fails. Stored link listings show expiry/revocation, never recoverable bearer tokens. Create a new link if lost.

The new current-fiction adapter stores only a SHA-256 capability hash. Its token travels in a URL fragment and an Authorization header, not a query/path or request log. The unauthenticated adapter serves only the immutable reader-safe document; all owner edition/assets/job APIs remain behind owner/origin/CSRF guards. Reader responses and pages use no-store/noindex/no-referrer, render text safely and make no AI call. Retired authoring and legacy share APIs remain disabled in production. This feature supplies no automatic hosting/deployment or promise of public reachability.

Schema 29 is appended without altering earlier migration source checksums. Source snapshots embed bounded art bytes in one synchronous read cut, independent of later media removal. Initial limits: 96 MiB per source, 1,000 sources / 512 MiB aggregate, 20 queued/running jobs, 10,000 retained jobs, 256 MiB per output and 2 GiB available rendered files. GC marks unreferenced files unavailable before unlink; new pins require availability. No story prose, private state or provider spend is rewritten.

03 B6 listening to saved fiction

Desk and Gallery offer **Listen to saved text**. Choose the configured MP3-capable Narrator voice, Read aloud, Auto-read or an audiobook edition, and a saved passage, path, chapter or volume. **Preview listening plan (free)** freezes that selection and shows segment counts, cached reuse, maximum synthesis calls and exposure. Only selected narrative prose goes to the Narrator: no Ask, private references, owner directions, correction records or sibling paths. Generated prose itself may contain fictional secrets; this is not semantic spoiler filtering.

A new purchase needs consent scoped to that immutable plan. Synthesis is sequential, with no transport retry, recursive splitting or purchased PCM fallback. Read aloud caps at 16 segments; Auto-read/audiobooks at 512. A unit caps at 16 MiB and an edition at 512 MiB. Choose a smaller passage/chapter/volume if preflight rejects the range. There is a 20-job installation queue and a conservative 4 GiB retained-audio quota.

The shared player keeps a labelled source, preparation progress, cancellation and Return action across rooms. Pause/Stop playback differs from **Cancel remaining synthesis**. Voice choice and playback speed are device preferences; changing speed and replaying a cached edition are free. A story/path change pauses Auto-read and requests cancellation of its remaining synthesis. A failed cancellation is reported rather than treated as confirmation. Tap to resume handles browser autoplay refusal without another purchase. Lock clears private playback UI.

Cancellation prevents undispatched units; dispatched work may still cost money and produce reusable cache bytes. Restart interrupts unfinished work without automatic resumption. Review a new plan to reuse completed units and purchase only permitted missing units. A disappeared or corrupt reviewed cache hit fails instead of silently buying a replacement. Retained job status shows known dollars and unknown-cost attempts separately. A late status reply cannot revive a cancelled player display.

Audio editions retain their source and are labelled older when the selected path has moved on. Download the MP3 separately. Private playable saves carry aggregate spend but exclude synthesized audio, listening plans/jobs, credentials and resumable purchase authority. Keep cold installation backups for configured database/media.

04 B5 retained visual collection

Gallery now lists the story-owned image collection in metadata pages of forty, including unplaced uploads and painting results. Thumbnails load lazily; Preview shows the full image, provenance and bounded retained-use metadata. Choose a cover, private reference or ancestral passage before placing an existing image. Placement and removal are local actions; earlier path snapshots and retained collection bytes remain available. New painting is one explicitly reviewed Illustrator purchase. A changed path, cancellation or restart cannot silently attach a late result: validated bytes remain ready and unplaced, and known/unknown spend remains visible. There is no automatic retry or reference-image conditioning.

The existing 400-image story ceiling remains. Every inventory member is an owned retained root; this batch exposes no permanent asset deletion or garbage collection. Playable save v2 carries all owned images and optional inventory descriptions and provenance, but no painting jobs, action keys, consent or provider credentials. Import creates new image identities and a fresh inventory concurrency value.

Visual Library offers selected-reference bundles (`.inkrefs`): worlds, characters and Scribes with normalized images and private lore/motives. Preview names matching existing entries; import always makes independent copies. A repeated identical import action returns the original copies, changed bytes conflict, and deleted results are reported rather than recreated. Bundles omit credentials, purchase journals and image-conditioning permission. They are unencrypted private files, not reader-safe books or full installation backups.

05 B4 editorial verification

Read recovery/B4-DELIVERY.md. Public version remains 5.0.0; B4 introduced schema 26 in an immutable migration module; current schema is 29. Preserve migration 24/25 and frozen save-v1 bytes. Genuine synthetic schema-25/B3-save fixtures were captured before B4; compare ledger, initial-state and beat bytes and inject rollback/reopen failure during migration 26. Verify current-format adoption and save-copy. Older fixtures remain historical evidence, not a backwards-compatibility obligation.

Tests must cover virtual episode defaults, immutable adoption/replay/fences, chronological non-overlap, ancestral narrative anchors, future/sibling rejection, empty/gap fallbacks, open-tail extension, fork-before/after behavior, incomplete private drafts, save closure/remapping and shared PublicationDocument grouping. Large automatic layouts must retain book behavior above adopted-record limits. Frozen documents retain previous headings. Empty plans never manufacture prose.

Browser journeys use a synthetic five-passage garden save created with injected responses, never paid provider calls. Exercise splits/volume assignment/preview, short keyboard viewport, exports, private draft reload/Lock, read-only historical outlines, restore preview and Back during a delayed load. Keep real Android/IME/ rotation and literary/deployment acceptance separate. Run full backend/frontend, desktop/mobile E2E and all four dependency audits; require all five exact-head CI checks before main merge. Source manuals and continuation state go on main. PDFs are regenerated together at this final recovery checkpoint.

06 B3 verification and documentation policy

Read recovery/B3-DELIVERY.md and CONTINUATION.md. Current storage is schema 29; public version remains 5.0.0. Never modify published migration 24 or frozen save-v1 modules. The genuine synthetic schema-24 database/baseline and B2-v2 save were captured before B3 implementation; keep them alongside original schema-23/v1 fixtures. Verify unchanged ledger/beat bytes, interrupted migration rollback/reopen, strict revision ancestry/producers, bound facts outside compaction and portable source closure. New response tests cover exact conditional Say, goal dispositions, editorial wishes, scene proposals and promise-cardinality-independent quiet scenes.

All possible quality stages must fit before the first provider call. Tests declare synthetic capacities explicitly; do not weaken unknown-capacity production checks to satisfy fixtures. Exercise plan expiry/change/replay, actual stage/candidate bounds, fixed reasoning allowance and known/unknown usage with deterministic transports. Browser tests use isolated authenticated data and mock paid endpoints; run desktop and mobile separately. Screenshots and emulation are not actual Android IME/rotation or literary acceptance, and authorize no paid benchmark/deployment.

The temporary feature-batch PDF deferral ends at this checkpoint. All six sources, rendered PDFs and the combined freshness manifest are updated together. Inspect every page through raster/contact-sheet and text/outline checks; the User Guide remains 35 A4 pages without shrinking its type or removing task coverage. Require all five checks green on the exact feature head before merging main, keep guides/state on main and delete only the integrated feature branch.

07 Final capacity evidence

`node scripts/benchmark-recovery.cjs` creates a private synthetic current-format store on the checkout's filesystem, measures 500/3,000/10,000 moments, and writes `docs/recovery/evidence/B8-CAPACITY.json`. It includes two paths sharing ancestry, four normalized images, sixteen reference revisions and dense evidenced facts. Queries have 30 warm samples; narration-context compilation has ten. Source file digests identify measured code. No provider or owner installation is opened.

The 500- and 3,000-moment saves round-trip both paths and source media. The dense 10,000-moment specimen exceeds the portable byte ceiling and fails explicitly before export; history is not dropped. Cold installation backup remains the way to retain a complete oversized store. Host metrics are not tablet/network or live literary acceptance. Use a scratch volume with adequate space; SQLite may already roll back a storage failure, whose original error must remain visible for diagnosis.

08 Repository and product ownership

Read AGENTS.md and the current release decisions before changing behaviour. The approved 5.0 product defaults to a reader-director outside the cast, excludes manual prose authoring and does not promise 4.x data migration.

The authoritative project checkout is under WSL at /home/rthorman/src/ink-morrow-5; shared Git metadata lives under /home/rthorman/src/ink-morrow/.git. Windows tools may work through WSL paths. Do not create a second source copy or recover abandoned older branches.

backend/src/app.js is the composer. Current game code lives in modules/fiction; the frontend starts its fiction app, not the old room router. Auth, provider, media and publication infrastructure is reused. legacy-runtime.js is opt-in for inherited tests, never enabled by server.js.

Treat existing user changes as owned work. Preserve unrelated edits, use a coherent feature branch and inspect the diff before committing. A failed command or CI rerun does not authorise history rewriting, deployment or data cleanup.

Project-owned 5.0 material remains AGPL-3.0-only. Preserve third-party licenses, bundled-font notices and the historical development credits. Do not describe generated fixtures as human playtesting or an unrun benchmark as research evidence.

B1 introduces five current-fiction room adapters and a contribution sheet using the shared dialog stack. Test preserved workflows through their new destinations: history/path/episodes in Chronicle, cast/facts/preferences in Codex, upload/paint in Gallery, and book/save export in Gate. Old `#/story/:id` links remain Desk aliases. B2 deepens Chronicle and Codex with server-bounded historical inspection and durable private drafts; B5 supplies complete retained image inventory. Do not revive authoring APIs. Retain historical schema/save fixtures as immutable evidence; current runtime/formats are the support target and backwards compatibility is waived; every schema change appends a checksum-bound migration. Verify rollback/reopen, exact quote/hash/remapping, public/private and sibling/future boundaries, cursor binding, draft conflicts, Lock and prepared-action recovery. `node scripts/benchmark-inspection.cjs` measures host query latency at 500, 3,000 and 10,000 moments; it is not Android or literary acceptance.

Regression cases include cross-room reading return without additional generation, an old bookmark outside the loaded window, foreign/deleted bookmark rejection, pending generation during history browsing, newer identical draft text, paid-review cancellation back to the same sheet, immediate Lock and reading reflow. Keep source/CI evidence separate from real Android IME/Back/rotation and stakeholder acceptance. All six source manuals are updated together; all six PDFs are regenerated at the final recovery checkpoint.

The PDF renderer serves only local documentation/type/brand assets on an ephemeral loopback HTTP listener, so it works in browser environments that prohibit file URLs. It never starts the application or opens its database. The listener closes when rendering finishes or fails.

09 Plan a substantial change

Choose a coherent user outcome and identify its state, UI, privacy, cost and documentation boundaries before implementation. Batches should be large enough to justify a full push/PR/CI/merge cycle, without combining unrelated risks.

For each mutation, name the input authority, expected revision, transaction, failure outcome and immutable record. For each provider call, name the role, data sent, maximum calls, consent scope, timeout, stale checks and known/unknown billing. For each read, name which path and visibility it can expose.

Retain existing deterministic invariants unless the owner explicitly changes them. New game design can challenge branding or compatibility, but that does not waive security, credential privacy, paid consent or data integrity.

Tests should accompany the behavioural change, not merely assert the new DOM. Include invalid input, delayed responses, navigation/lock, concurrent revision, restart and zero-provider paths where relevant. Use fixtures with exact expected effects rather than accepting whichever prose a model returns.

Update all affected manual sources; regenerate the six-book PDF set at final recovery completion. A substantial feature is incomplete if the user guide still describes retired controls or if a version label masks inherited operating instructions.

10 Local verification layers

From the repository root, run npm run lint, npm run test:setup, npm run check:release and npm run check:brand. Backend Jest runs through node node_modules/jest/bin/jest.js in backend. Frontend Jest uses node --experimental-vm-modules node_modules/jest/bin/jest.js in frontend.

Backend tests use isolated databases and injected providers. New production boundary tests instantiate the default composer without legacyEnabled. Inherited component tests may explicitly opt into the old runtime; their coverage is regression protection, not a claim of a live 5.0 feature.

Browser tests use port 3100, an in-memory database, fresh media and an isolated setup credential. Never point them at the owner's port 3000 or real database. Sweep abandoned E2E servers using the supplied script; do not kill processes by a broad command-line substring.

Run desktop Chromium and Mobile Chrome in separate invocations so each receives a fresh server. Use an existing browser-capable environment; do not download a browser merely to satisfy local verification. CI has its own browser setup.

A passing test count is evidence about that exact code/configuration. Record full-suite versus added focused cases accurately. Do not imply a paid provider, different browser engine or human session was tested when only mocks ran.

11 Data and concurrency qualification

The current family is ink-morrow-5, schema 29. Fresh creation, supported migration, repeat startup, rollback, ledger integrity, future-family refusal and source-file preservation need independent tests. Never edit a migration checksum merely to make a valuable database open.

Exercise old database files with WAL and missing shared-memory companions. Preflight must inspect a private copy, preserve original bytes/timestamps and not create source sidecars. Test orphan journals, symlinks, failed copy/storage and source changes. A current valid WAL-only committed state must recover.

For paid work, delay the provider, mutate or restart, then return the response. Assert no stale canon, complete known/unknown billing and no duplicate purchase. Quality requires per-call records and one shared repair allowance. Transport retry is disabled even on uncertain failure.

For graph changes, test exact fork snapshots, sibling exclusion, old fact retrieval beyond the working set, corrected/retired versions and evidence remapping through save import. Test all references before any import write.

Use long deterministic histories to test bounded readers and retrieval. A large fixture proves structural scale and invariants, not that an actual model will remember every relevant detail or remain enjoyable across a long campaign.

12 UI, accessibility and one logo

Immediate feedback is part of correctness. Dialogs should open before slow supporting requests; buttons must show reviewing/submitting status and prevent duplicate work. Cancelling, failing or changing routes should preserve or discard drafts according to the explicit lifecycle, never by accident.

Exercise keyboard navigation, focus restoration, labelled controls, mobile reflow, delayed loads and lock while a request is pending. Automated accessibility scans supplement real inspection; they do not prove a design is easy to use.

The canonical product lockup is frontend/brand/ink-morrow-lockup.svg, the existing artwork at the top of GitHub README. App header/footer, auth threshold and all manual covers/running headers reuse it. Do not regenerate it or emulate its lettering with another font. Ordinary product-name mentions remain text.

npm run check:brand includes a canonical-logo reference guard. Browser tests verify the loaded image in the reader and locked threshold. Inspect both narrow and wide screenshots for size and clipping; inspect rendered PDF covers and headers, not only SVG paths in source.

Images above manuscript text and separate preceding EPUB image pages are distinct requirements. Check every EPUB spine item and reader layout rather than assuming shared HTML produces both representations correctly.

13 The six-book documentation workflow

Active books are declared in docs/pdf-library/books.mjs and use Markdown sources with a shared renderer/theme. The previous fixed HTML guide remains a historical content and visual baseline, not a current workflow.

Update the User Guide, Operations, Architecture, State Machine Atlas, Security and Maintainer books where affected, then run npm run docs:pdf. All six PDFs and generated.json form one artifact set. The active filenames, cover labels, metadata and headers use 5.0.

Run strict freshness after rendering. PDF QA checks extractable text, replacement characters, current identity, outlines and page counts. Rasterize pages and inspect covers, tables, code, headers and final pages. A valid PDF hash cannot prove layout or absence of a clipped paragraph.

The guide includes complete journeys using no optional provider features, the whole current feature set and coherent subsets. No-provider play means existing reading/local management, not manual authoring or an invented offline narrator. Every journey identifies costs, skipped features and its save/book outcome.

Retain historical release evidence, but make current entry-point documentation unambiguous. Never claim a book is current merely because its filename changed. A changed source or brand asset requires regenerating the complete set.

14 CI, release and exact-head merge

The five CI gates are Brand residue, Lint, Jest, Production dependency audit and Playwright E2E. The brand job also checks release identity and warns about PDF freshness. For release work, run strict freshness locally even though the general CI freshness check is advisory.

Feature PRs target release/5.0.0. Push a coherent batch, inspect the PR diff and wait for all five gates on its exact current head. A green earlier commit is not authority to merge a changed head. Use a head-matching merge and verify the resulting merge commit.

After each feature merge, update from the integration head before the next batch. Do not rewrite the historical 4.x release line. Unrelated dependency PRs are not part of this product programme merely because they are open.

Final release-to-main integration is authorised only after all approved 5.0 work is complete and the final PR's current head is green. Record actual PR links, head/merge hashes, tests and remaining limitations in the release record. Do not label planned checks as passed.

Merging code is not deployment. The owner requested the old port-3000 instance stopped; do not restart or replace it without a separate request. A final handoff should briefly distinguish code state, running state and 4.x compatibility.

15 Definition of done and regression response

A completed change has a verified user path, bounded state transition, correct privacy/cost boundary, failure and race coverage, current documentation and an honest release record. For model-facing changes, distinguish protocol tests from semantic evaluation and do not promise unmeasured quality gains.

A completed 5.0 release additionally has fresh-family storage isolation, no live manual-authoring or retired room APIs, consistent branding, complete generated manuals and green exact-head integration. Old user data remains untouched. Character/template portability is not a release blocker and is not advertised unless implemented.

When a regression appears, reproduce it with minimal fixture data. Determine whether the defect is product code, a stale test assumption, environment, provider behaviour or external infrastructure. Do not weaken an invariant simply to restore a green badge.

Fix a proven defect in a focused but substantial batch, rerun the affected layer and full release gates in proportion to risk, and record the evidence. An intermittent registry error may justify a CI rerun; it does not justify bypassing the audit.

If completion needs a new product choice, paid live evaluation, deployment authority or destructive recovery, pause and ask. Persistence toward a merge does not broaden the authorised actions. Keep the owner informed with concise, meaningful updates rather than unchanged polling reports.

16 Optional quality verification

Fixture tests cover Off/Standard/Memory/Both, first-pass acceptance, the sole structural or consistency repair, replacement re-review and terminal rejection. Assert maximum totals of one/four/six, role-specific model routing and disabled transport retries. Reject malformed or unevidenced review JSON; approval must not bypass deterministic ownership, fourth-wall, state-effect or adjudication checks. No canon mutation may happen between draft and final acceptance.

Test missing/stale plan identity, unavailable memory before the first purchase, provider changes after an already billed draft, mixed known/unknown costs, restart and late results, zero-call unchanged rulings and free idempotency replay. Rewind/save tests preserve quality choices and aggregate spend without carrying request authority. UI tests require fresh scoped review despite old global consent, cancelled-direction retention, free progress reads and lock/navigation fencing. Exercise the real preference API and both desktop/mobile browser journeys.

These automated protocol fixtures make no paid provider requests and establish no model-quality ranking. Live comparisons need separate authorisation and versioned results: model/provider/date, repeated-persuasion and justified-cooperation cases, knowledge/ownership/continuity errors, review false positives, latency and actual known/unknown cost. Never claim the extra calls guarantee better play.

17 People and episode verification

Exercise both authored openings from development to payoff and aftermath, with Follow/Steer sufficient and no compulsory avatar. The garden fixture covers a refused group role, a free unchanged repeat and cooperation on new agreed terms. The mystery fixture distinguishes affection from restored practical trust. These are protocol fixtures, not evidence from human playtesting or paid models.

Test relationship aspects/targets, exact evidence, owned-character boundaries, rejection of numeric meters and world-fact rewriting, historical provenance, reload/fork/save restoration and remapped payoff identities. Test quiet scenes, clarification and local correction cannot manufacture a played payoff. Ending early remains possible; goal completion never forces an episode end.

Recaps must find narrated moments despite many local changes, filter public commitments/relationships before limits, exclude retired and other-path records, and make zero provider calls. Browser tests cover immediate recap feedback, episode questions, reload, narrow reflow and retained path/episode drafts after failed writes. Include optional standard/memory/both consistency and fresh-storage isolation in the same final integration regression evidence.

18 Fourth-wall verification

Test default Never, invalid choices, Living-world-only UI visibility, local preference saves and restored settings. Cover optional consecutive Freely addresses, the five intervening narrated scenes required by Rarely, and no cooldown advancement on failure or clarification. Reject unknown/inhabited speakers and effects evidenced only by an aside. Verify known charges survive rejection and, with quality Off, no automatic extra request occurs.

Reload, fork and save-copy must preserve branch-local settings and counters; reject future scene indices in imported saves. Addresses remain ordinary saved passage text for publication. Mocked protocol tests do not demonstrate semantic model compliance with Never or prove resistance quality. The optional consistency pipeline has its own verification contract above; fourth-wall permission alone does not enable it or authorise another model call.

19 Clear-influence verification

Cover the default one-moment direction, explicit ongoing focus, failure retention, local focus release, rewind and save-copy restoration. Invitations must not submit, overwrite drafts, reveal hidden facts or supply an inhabited person's decisions. Challenge preflight and repeat must make zero provider calls; stale preflight must not bypass paid consent. Test navigation during review as well as generation.

Exercise recall beyond 128 working facts, public filtering before the 32-result limit, retirement and sibling-path rejection. Changed-fact citations must point to real earlier records, and portable copies must remap them. Exercise shelf paging beyond 200 stories, immediate loading feedback, keyboard operation and 320-pixel reflow. Automated accessibility scans supplement, not replace, human usability testing. All six books remain one regenerated artifact set per feature PR.

20 Fair-resistance verification

The deterministic royal-guard fixtures test twenty repeated/paraphrased requests without extra purchases, legitimate new recorded authority, unknown or hidden evidence, distinct styles, narrator outcome disagreement, private-field filtering, branch restoration and save import. Memory fixtures exceed 128 facts, retrieve an old relevant fact, correct and retire it, and exclude a sibling path's history. Run `node node_modules/jest/bin/jest.js --runInBand tests/fiction-resistance.test.js` from backend. These use mocked providers and do not establish model suitability.

Live evaluations need an explicitly approved paid budget and repeatable fixtures. Record exact model/settings/date, latency, cost, contradictions, unjustified capitulation AND unjustified stubbornness. Human semantic review is necessary; another model's approval is not sufficient. No model is certified by unit tests. The owner approved the three substantial iteration PRs and final green integration to main. Deployment remains separate; the old port-3000 instance is now stopped.

21 Illustrated-save verification

Keep the media/save batch substantial. Test synchronous dialog feedback, prevented duplicate image purchases, retained drafts on failure, late-response route/lock isolation, exact path-local placements, safe upload and local description correction. Test EPUB image-only spine resources before associated prose, while manuscript/HTML places the image above that prose. Re-read the whole EPUB spine, not only book.xhtml.

Save tests must round-trip hidden truth, knowledge, commitments, resources, control, episodes, preferences, director history, all branches, illustrations and known/unknown spend. Reject malformed versions/fields, cycles, dangling/future/cross-path references, unsafe media and corrupt digests before writes. Inject import transaction failure and verify both database rollback and cleanup of only newly staged image files. Browser journeys must upload, export, download a private save and import a copy. Never use real provider spending to satisfy deterministic tests. Verify final 5.0 identity, fresh storage and the complete current documentation together.

Visual catalogue coverage must exercise world, character and Scribe CRUD and both upload/generation paths, plus story covers and all reference portraits. Verify auth/CSRF before uploads, one pending purchase, changed-provider rejection, failed and late known/unknown charges, safe normalization and owned-file cleanup. Ordinary details, selection and upload make no model request. Old authoring routes stay 404.

Reference-development coverage must exercise all three typed output shapes, empty and partial seeds, private-field disclosure, exact provider/model review, idempotent replay, one invalid-output correction, combined known/unknown spend and restart interruption. Browser coverage proves the AI result fills an editable unsaved studio and that Save & add image moves to the explicit upload/painting boundary without a hidden purchase.

Freeze and copy selected images at setup. Delete or change the catalogue, then check that the story and its images remain unchanged. Round-trip private references and all visual asset IDs through saves; reject dangling or malformed copies. Books must contain the cover and passage art, never private lore or reference portraits. Test schema-22 to 23 migration with identical earlier ledger rows. UI tests cover immediate dialogs, retained selection/drafts, bounded tablet images, catalogue pagination, lock clearing and stale responses. Use mocked model output, not live spend.